

Md Motaleb Hossen Manik

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Research Statement

I am currently a **Ph.D. student in Computer Science** at **Rensselaer Polytechnic Institute**, where my research centers on **multimodal AI for safety-critical domains**. My work focuses particularly on medical imaging education, workflow auditability, and synthetic agent societies, combining benchmark design, interactive system development, and risk-aware evaluation with emphasis on reproducibility, provenance, and accountability. I am seeking a **tenure-track faculty position** or **research scientist role** to advance large language models, multimodal AI, medical imaging, and agentic systems in trustworthy, interpretable, and human-centered directions.

Academic Credentials

Ph.D. in Computer Science

Aug 2024 – Present

Rensselaer Polytechnic Institute (RPI), Troy, NY 12180, USA

Advisor: Prof. Ge Wang, Department of Biomedical Engineering, with affiliation in Computer Science, RPI

Research Group: Wang-AXIS Lab, RPI

M.Sc. Eng. in Computer Science and Engineering

Sep 2020 – Mar 2023

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 4.00/4.00

Advisor: Prof. Dr. Kazi Md. Rokibul Alam, Department of Computer Science and Engineering, KUET

Thesis: *A Study on a Blockchain-Based Secure Framework for Multi-party Skyline Queries*

B.Sc. Eng. in Computer Science and Engineering

Nov 2015 – Mar 2020

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 3.93/4.00

Rank: 1st out of 134 graduates

Advisor: Prof. Dr. M. M. A. Hashem, Department of Computer Science and Engineering, KUET

Thesis: *A Study on Sentiment Analysis of Bangla and Phonetic Bangla Reviews*

Teaching Experience

Teaching Assistant

Aug 2024 – Present

Department of Computer Science, Rensselaer Polytechnic Institute (RPI), Troy, NY

Courses:

- CSCI 2600 *Principles of Software* (Spring 2025, Summer 2025, Spring 2026)
- CSCI 4430 *Programming Languages* (Fall 2025)
- CSCI 4440 *Software Design and Documentation* (Summer 2025)
- CSCI 1700/2700/4968/4970 *RCOS* (Fall 2024, Spring 2025), including evaluation of student open-source project presentations

Assistant Professor (On Study Leave)

Jun 2024 – Present

Department of Computer Science and Engineering, KUET, Bangladesh

Department of Computer Science and Engineering, KUET, Bangladesh

Courses:

- CSE 2103/2104 *Microprocessors and Microcontrollers* (theory and lab)
- CSE 3206 *Robotics Laboratory* (lab)
- CSE 3217 *Mobile Computing* (theory)
- CSE 4120 *Technical Writing and Seminar* (lab)
- CSE 4223/4224 *Digital System Design* (theory and lab)

Research Experience

Multimodal AI, LLMs & Medical Imaging Education

2019 – Present

Ph.D. at RPI; earlier ML/NLP foundation

My recent research focuses on multimodal AI systems for medical imaging education, benchmark design, and interactive learning. Built **MEDI-SLATE**, a slide-lecture aligned dataset containing **1,117** high-resolution slides and **262,182** refined narration tokens from a full **23-lecture** undergraduate medical imaging course. Designed **MILU**, a benchmark for structured lecture understanding across four open-source VLMs, generating **15,000+** JSON artifacts in **9–11 GPU-hours** on $4 \times$ **NVIDIA RTX A5000** GPUs; across **1,117** slides, parsing coverage remained **92–99%**, while semantic agreement stayed low, with pairwise concept Jaccard of **0.03–0.09** and triple-level F1 of **0.001–0.033**. Contributed to **ALIVE**, a fully local avatar-lecture interaction engine integrating ASR, FAISS-based retrieval, local LLM reasoning, text-to-speech, and talking head synthesis for real-time lecture-grounded interaction. Earlier work in ML and NLP included Bangla and phonetic Bangla text analysis, where a manually annotated dataset of **1,500** reviews achieved **75.58%** accuracy with **SVM**, and later work on unified sentiment and emotion recognition.

AI Safety & Synthetic Agent Systems

2025 – Present

Ph.D., RPI

My current safety-oriented work studies risky instruction propagation, social regulation, and decentralized governance in synthetic AI societies. Developed **OpenClaw on Moltbook**, an agent-only social environment for analyzing emergent behavior among autonomous agents. In an empirical study of **39,026** posts and **5,712** comments from **14,490** agents, found that **18.4%** of posts contained action-inducing language and that such posts were more likely to elicit norm-enforcing responses, while toxic responses remained rare. Also developed **ADAPT**, an AI-driven decentralized publishing framework that models scholarly publishing as a **closed-loop governance system** with bounded policy adaptation under overload, disagreement, and collusion-related stress; this work led to **U.S. Provisional Patent Application No. 63/975,609**.

Blockchain Provenance, Secure Retrieval & Applied AI

2020 – Present

M.Sc. at KUET + continuing work

My M.Sc. and related work focus on blockchain based trust, provenance, and secure information systems. Completed my M.Sc. thesis on a blockchain based secure framework for user-centric multi-party skyline queries, introducing **multi-party ElGamal, re-encryption and shuffling, targeted queries**, and blockchain based integrity with distinct blocks for each party; the framework reduced the reported skyline computation from $(L \times M \times N)$ to $(L + M) \times N$. Built **SlideChain**, a blockchain-backed semantic provenance framework for educational AI, using **four** VLMs over **1,117** lecture slides and achieving approximately **one-slide-per-second** registration throughput, **100% tamper detection**, and deterministic reproducibility with **Jaccard = 1.0**. Also developed secure data-sharing and applied AI systems including **ShaEr**, a privacy-preserving medical data sharing and monetisation framework, and contributed to a blockchain-aided heart disease detection system integrating **seven** datasets and achieving **89.2%** accuracy, with **85.3%** precision, **97.0%** recall, and **90.8%** F1 using a voting ensemble with private blockchain support.

Journal Articles

2026

- **Manik, M. H.**, and Wang, G. “Evaluating Risk and Auditability in Workflow Agents for Safety-Critical Domains.” *Journal of Information and Intelligence*, Accepted, 2026.
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “MEDI-SLATE: A Slide–Lecture Aligned Text Ensemble for Medical Imaging Education.” *Visual Computing for Industry, Biomedicine, and Art*, in press. [IF 6.0; ESCI; Springer] Link
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “MILU: A Consensus Ensemble Benchmark for Multimodal Medical Imaging Lecture Understanding.” *Journal of Medical Imaging*, 13(6):062202, 2026. [IF 1.7; Q2; SCIE/Scopus; SPIE] Link

2025

- **Manik, M. M. H.**, Muldowney, W., Islam, M. Z., and Wang, G. “Development of an Optically Emulated CT (OECT) Scanner for College Education.” *Visual Computing for Industry, Biomedicine, and Art*, 2025. [IF 6.0; ESCI; Springer] Link
- Ahmed, M. R., Aziz, A., **Manik, M. M. H.**, and Habib, M. A. “Blockchain-aided Comparative Study of Heart Disease Detection Using Machine Learning-Based Approaches with an Expanded Dataset.” *Computers in Biology and Medicine*, 2025. [IF 7.0; Q1; SCIE; Elsevier] Link
- Habib, M. A., and **Manik, M. M. H.** “ShaEr: A Blockchain-Based Framework for Secure Medical Data Sharing and Monetisation.” *IET Blockchain*, 2025. [Q2; Scopus; Wiley/IET] Link

2023

- **Manik, M. M. H.** “A Novel Approach in Determining Areas to Lockdown During a Pandemic: COVID-19 as a Case Study.” *International Journal of Information Engineering and Electronic Business*, 15(2):30, 2023. Link

2021

- Rahman, M. M., Islam, M., **Manik, M. M. H.**, Islam, M. R., and Al-Rakhami, M. S. “Machine Learning Approaches for Tackling Novel Coronavirus (COVID-19) Pandemic.” *SN Computer Science*, 2:1–10, 2021. [IF 2.38; Q2; Scopus; Springer] Link

Conference Papers

2025

- **Manik, M. M. H.**, Sagor, A. H., Mondal, F. A., Touhid, M. M., and Islam, M. Z. “Unifying Sentiment Analysis and Emotion Recognition for Bangla Text: A Hybrid Approach.” *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025. Link

2024

- Atik, A. I., Rahman, A., Sami, S. A., and **Manik, M. M. H.** “An Android Application for Healthcare Management System with ML-Driven Solution.” *27th International Conference on Computer and Information Technology (ICCIT)*, 2024. Link
- Saha, A., Majumder, S., **Manik, M. M. H.**, and Hashem, M. A. “Enhancing Hyperledger Fabric: A Scalable Framework for Optimized Blockchain Performance.” *27th International Conference on Computer and Information Technology (ICCIT)*, 2024. Link
- Islam, M. R., Alam, K. M. R., and **Manik, M. M. H.** “Decentralized GDPR Compliance: A Blockchain Framework for Personal Data Management.” *27th International Conference on Computer and Information*

Technology (ICCIT), 2024. Link

2023

- **Manik, M. M. H.**, Mahadi, J., Touhid, M. M., and Alam, K. M. R. “Enhancing Robustness of Machine Learning Algorithms for Bangla Text Classification: A Defensive Approach against Adversarial Attacks.” *6th International Conference on Electrical Information and Communication Technology (EICT)*, 2023. Link
- **Manik, M. M. H.**, Sagor, A. H., Habib, M. A., Touhid, M. M., Ahmed, T., Islam, M. Z., and Haque, F. “Redefining Crime Record Storage: An Advanced Architecture Harnessing the Power of Blockchain Technology.” *26th International Conference on Computer and Information Technology (ICCIT)*, 2023. Link
- Majumder, S., Zaha, R., **Manik, M. M. H.**, and Alam, K. M. R. “A Blockchain Based Scalable Framework for Academic Document Verification.” *26th International Conference on Computer and Information Technology (ICCIT)*, 2023. Link
- Habib, M. A., **Manik, M. M. H.**, and Zaman, S. “A Blockchain-Based Technique to Prevent Grade Tampering: A University Perspective.” *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2023. Link
- Habib, M. A., and **Manik, M. M. H.** “A Technique to Avoid Blockchain Denial of Service (BDoS) and Selfish Mining Attack.” *Fifth International Conference on Blockchain Computing and Applications (BCCA)*, 2023. Link

2022

- **Manik, M. M. H.**, Haque, F., Hashem, M. M. A., Habib, M. A., Islam, M. Z., and Ahmed, T. “A Hybrid Framework for Sentiment Analysis from Bangla Texts.” *25th International Conference on Computer and Information Technology (ICCIT)*, 2022. Link
- **Manik, M. M. H.**, Alam, K. M. R., and Morimoto, Y. “A Blockchain Based Secure Framework for User-Centric Multi-Party Skyline Queries.” *25th International Conference on Computer and Information Technology (ICCIT)*, 2022. Link
- **Manik, M. M. H.**, Habib, M. A., and Ahmed, T. “Machine Learning Algorithms on COVID-19 Prediction Using CpG Island and AT-CG Feature on Human Genomic Data.” *International Conference on Machine Intelligence and Emerging Technologies*, 2022. Link

2020

- Rahman, M. M., **Manik, M. M. H.**, Islam, M. M., Mahmud, S., and Kim, J. H. “An Automated System to Limit COVID-19 Using Facial Mask Detection in Smart City Network.” *IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS)*, 2020. Link

2019

- Haque, F., **Manik, M. M. H.**, and Hashem, M. M. A. “Opinion Mining from Bangla and Phonetic Bangla Reviews Using Vectorization Methods.” *4th International Conference on Electrical Information and Communication Technology (EICT)*, 2019. Link

Preprints

2026

- **Manik, M. M. H.**, and Wang, G. “HypothesisMed: Inference-Time Answer Fusion and Structured Hypothesis-Space Reporting for Biomedical Question Answering.” arXiv:2606.00971, 2026. Link
- **Manik, M. M. H.**, and Wang, G. “Unified Deployment-Aware Evaluation of Open Reasoning Language Models.” arXiv:2604.07035, 2026. Link
- **Manik, M. M. H.**, and Wang, G. “Emergent Decentralized Regulation in a Purely Synthetic Society.” arXiv:2604.06199, 2026. Link
- **Manik, M. M. H.**, and Wang, G. “OpenClaw Agents on Moltbook: Risky Instruction Sharing and Norm

Enforcement in an Agent-Only Social Network.” arXiv:2602.02625, 2026. Link

2025

- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “SlideChain: Semantic Provenance for Lecture Understanding via Blockchain Registration.” arXiv:2512.21684, 2025. Link
- Islam, M. Z., **Manik, M. M. H.**, and Wang, G. “ALIVE: An Avatar-Lecture Interactive Video Engine with Content-Aware Retrieval for Real-Time Interaction.” arXiv:2512.20858, 2025. Link
- **Manik, M. M. H.**, Islam, M. Z., and Wang, G. “N-ReLU: Zero-Mean Stochastic Extension of ReLU.” arXiv:2511.07559, 2025. Link
- **Manik, M. M. H.** “ChatGPT vs. DeepSeek: A Comparative Study on AI-Based Code Generation.” arXiv:2502.18467, 2025. Link

2024

- **Manik, M. M. H.**, Habib, M. A., Islam, M. Z., Ahmed, T., and Haque, F. “Analyzing the Dynamics of COVID-19 Lockdown Success: Insights from Regional Data and Public Health Measures.” arXiv:2402.18594, 2024. Link
- Roy, S. C., and **Manik, M. M. H.** “Question-Answering System for Bangla: Fine-Tuning BERT-Bangla for a Closed Domain.” arXiv:2410.03923, 2024. Link

2022

- Habib, M. A., **Manik, M. M. H.**, and Khulna, B. “Classification of DNA Sequence Using Machine Learning Techniques.” EasyChair preprint, 2022. Link

Submitted Manuscripts

- **Manik, M. M. H.**, Alam, K. M. R., and Morimoto, Y. “A Secure Framework for User-Centric Multiparty Skyline Queries via Pruned and Prioritized Datasets.” Submitted to *Future Generation Computer Systems*; under review. [IF 6.1; Q1; SCIE]
- Habib, M. A., and **Manik, M. M. H.** “Blockchain-Enabled Secure Land Record Management System with a New Lightweight Cryptosystem Based on Hybrid Chaotic Map.” Submitted to *Knowledge-Based Systems*; first revision in process. [IF 8.8; Q1; SCIE]
- **Manik, M. M. H.**, and Wang, G. “ADAPT: AI-Driven Decentralized Adaptive Publishing Transformer.” Submitted to *Research Integrity and Peer Review*; under review.
- **Manik, M. M. H.**, and Wang, G. “Emergent Decentralized Regulation in a Purely Synthetic Society.” Submitted to *Scientific Reports*; under review. [IF 3.8; Q1; SCIE; Nature Portfolio]

Honors & Awards

- **Best Paper Award**, International Conference on Electrical, Computer and Communication Engineering (ECCE), IEEE, 2023.
- **Dean’s Award** (three consecutive years), Faculty of Electrical and Electronic Engineering, KUET, awarded for top academic performance among all CSE students.
- **Merit Scholarship**, KUET (2016–2020), granted to students ranking in the top 5% of their cohort.

Grants & Funding

- **AI-Integrated Curriculum Development Award** (2026). Funded project: *AI-Integrated Interactive Classroom Platform (BMED 2300)*. Award amount: **\$5,000**. Funding period: **May 15–August 15, 2026**. PI: Prof. Ge Wang (RPI); Role: Graduate Research Collaborator.

- **NVIDIA Academic Grant Program** (2025). Hardware grant: $1 \times$ NVIDIA DGX Spark. Project: *Iterative Avatar Teaching in the NVIDIA Omniverse*. PI: Prof. Ge Wang (RPI); Role: Graduate Research Collaborator.

Patents

- **Manik, M. M. H.**, and Wang, G. “ADAPT: AI-Driven Decentralized Adaptive Publishing Testbed.” U.S. Provisional Patent Application No. 63/975,609, filed February 4, 2026. Rensselaer Polytechnic Institute.

Professional Service

Peer Reviewer — Journals

- **Engineering Reports**, Wiley, 2026.
- **Technology in Society**, Elsevier, 2025.
- **Journal of Systems and Software**, Elsevier, 2025.

Peer Reviewer — Conferences

- IEEE COMPAS 2025; EICT 2025; ICCAconf 2024; EICT 2023.

Mentorship & Student Supervision

- Supervised 3 undergraduate B.Sc. thesis groups and 2 undergraduate project groups at the Department of Computer Science and Engineering, KUET (2022–2024), covering topics in machine learning, blockchain systems, and large language models.

Invited Talks & Presentations

- “**All About AI: Avatarized Online Course**,” ECSE Best Practices Series, Department of Electrical, Computer and Systems Engineering, Rensselaer Polytechnic Institute, August 7, 2025. Invited by Prof. Shayla Sawyer (Director, Mercer XLab, RPI). Presented the ALIVE avatarized lecture system to a cross departmental faculty audience and received commendation from senior RPI faculty for its significance and experimental potential.

Technical Skills

Machine Learning, LLMs & Multimodal AI

Large language models, vision–language models, multimodal learning, transformer architectures, retrieval augmented generation, prompt engineering, benchmark design, model evaluation, scientific slide understanding, medical imaging AI.

Frameworks & Libraries

PyTorch, TensorFlow, HuggingFace Transformers, SentenceTransformers, Scikit-learn, NumPy, Pandas, Matplotlib, FAISS, Whisper ASR, local LLM pipelines, structured JSON evaluation workflows.

Agentic Systems & AI Safety

Synthetic agent societies, workflow agents, norm enforcement, action-inducing risk modeling, auditability, decentralized governance, trustworthy AI evaluation.

Blockchain, Security & Provenance

Ethereum, Solidity, Hyperledger Fabric, Hardhat, IPFS, smart contracts, cryptographic hashing, Keccak-256 commitments, decentralized storage, blockchain provenance, secure medical data sharing, research-integrity workflows.

Programming & Tools

Python, C/C++, Java, JavaScript, SQL; Git, Linux/Bash, Jupyter Notebook, REST APIs, JSON pipelines, L^AT_EX, reproducible experiment logging.